SPEAK

Language-Learning Platform for Primary-School Children

Software Requirements Specification

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Client: SPEAK Language-Learning Programme
Status: For Review and Approval

1. Introduction and System Overview

SPEAK is a web-based platform that helps primary-school children learn a language (German vocabulary and grammar) through short, guided, audio-first activities. Learning content is organised by week and by area (Vocabulary and Grammar) and is delivered as a sequence of small interactive tasks such as storybook listening, audio-to-image matching, picture selection, article and case exercises, and memory games. The platform is operated by teachers, who manage their learners, assign content, and review each child's progress, and it is administered by a system administrator, who manages teacher accounts and the learning content.

The existing learning material was originally authored as psychology-experiment activities (OpenSesame and OSWeb) and delivered through the JATOS research platform. This project rebuilds that material as a purpose-built web application. The interactive task engine and the management features are developed from scratch. The existing media assets (images and audio) and the task content (currently held in CSV files) are reused and imported into the new system.

1.1 Purpose

This document specifies the functional, non-functional, and technical requirements for the SPEAK platform. It establishes the agreed scope between the developer and the client before development begins. Each requirement is uniquely identified so that it can be referenced during design, implementation, and acceptance testing.

1.2 Scope

The system provides teacher registration and login; administrator management of teachers and learning content; teacher-managed student accounts with access codes; organisation of students into classes; assignment of weekly modules; an interactive, adaptive task player for children; recording of every learner response; and a teacher dashboard for reviewing progress and results. The first release covers the content already provided by the client (Weeks 6, 10, and 12; Vocabulary and Grammar). The content is delivered through a data-driven engine, so that further weeks can be added by importing content without modifying application code.

1.3 User Roles

The system supports three roles. All three are mandatory in the first release.

Role	Description
Administrator	Logs in with email and password. Manages teacher accounts (create, activate, deactivate), imports and updates learning content (weeks, tasks, media), and oversees system-wide configuration.
Teacher	Registers and logs in with email and password. Creates and manages student accounts and access codes, organises students into classes, assigns weekly modules, and reviews individual and class progress.
Student (Child)	Logs in with an access code issued by the teacher. No email address or self-chosen password is used. Works through assigned modules by listening, tapping, and answering, and receives immediate feedback.

1.4 Task-Type Glossary

The content supplied by the client contains the following recurring task types. They are listed here because several functional requirements refer to them directly.

Code	Task type	What the child does
A	Storybook	Listens to a narrated story that introduces target words with definitions.
B	Audio to image	Hears a spoken word and selects the matching image from several options.
F_1, F_2	Audio choice	Sees an image and chooses the correct spoken word from two audio options, one of which is a sound-alike distractor.
T	Picture choice	Hears a target word and chooses the correct picture from two options.
R	Yes / No reflection	Answers closed yes/no questions about target words, supported by an image.
G1, G2, V_3	Grammar and articles	Works with grammatical cases (nominative, accusative, dative) through rhymes, article selection, and gap-fill sentences.
U	Memory (suitcase)	Memorises named objects, then recalls which of them were shown.
AT, AV, AX	Teaching and reflection	Guided screens that teach how articles change across cases, with picture support.
C (adaption)	Remediation	Additional practice presented only to a child who exceeded the mistake threshold in the preceding task.

2. Functional Requirements (Must)

2.1 Accounts and Authentication

R01: The system must provide registration for teachers, capturing name, email address, and password.

R02: The system must provide email-and-password authentication and login for teachers.

R03: The system must provide email-and-password authentication and login for administrators.

R04: The system must allow a child to log in using an access code only, without an email address or a self-chosen password.

R05: The system must allow a teacher to create, edit, and deactivate student accounts.

R06: The system must generate a unique access code for each student account and allow the teacher to regenerate that code.

R07: The system must allow a teacher to organise students into classes.

R08: The system must allow an administrator to create, activate, and deactivate teacher accounts.

R09: The system must allow an administrator to import and update learning content, comprising modules, tasks, trials, and media assets.

R10: The system must allow a user to log out, which invalidates the current session on the client.

2.2 Access Control (Route Guards and Role Guards)

Access control is enforced on both the client and the server. The client prevents unauthorised navigation; the server independently verifies every request, so that access control cannot be bypassed by manipulating the browser.

R11: The system must implement a route guard on the frontend that redirects any unauthenticated user who requests a protected route to the appropriate login page.

R12: The system must implement a role guard on the frontend that permits access to a route only if the authenticated user holds the role required by that route (Administrator, Teacher, or Student).

R13: The system must display only the navigation items permitted for the authenticated user's role.

R14: The system must verify the session token and the user's role on the server for every protected API request, and reject any request whose token is missing, invalid, or expired.

R15: The system must reject any request in which an authenticated user attempts to access data belonging to another user, class, or teacher.

R16: The system must return an explicit authorisation error and no protected data when a role guard or a route guard denies access.

Explanation: *Route guards restrict access on the basis of authentication (is the user logged in). Role guards restrict access on the basis of role (is this user permitted to perform this action). Requirements R11 to R13 cover the frontend implementation; requirements R14 to R16 cover the server-side enforcement that backs it.*

2.3 Content and Assignment

R17: The system must present learning content organised by week and by area (Vocabulary and Grammar).

R18: The system must deliver a module, defined as one week within one area, as an ordered sequence of tasks.

R19: The system must allow a teacher to assign a module to a class or to an individual student.

R20: The system must load task content and media from the imported content store, which is generated from the CSV files and media assets supplied by the client.

2.4 Interactive Tasks

R21: The system must provide a storybook listening task with narrated audio and word definitions (type A).

R22: The system must provide an audio-to-image selection task, in which the child hears a word and chooses the matching image (type B).

R23: The system must provide an image-to-audio selection task with a sound-alike distractor (types F_1 and F_2).

R24: The system must provide a picture-choice task, in which the child hears a target word and chooses the correct picture (type T).

R25: The system must provide a yes/no reflection task about target words, with image support (type R).

R26: The system must provide grammar tasks covering grammatical cases, comprising article selection and gap-fill sentences (types G1, G2, and V_3).

R27: The system must provide a memory (suitcase) task (type U).

R28: The system must provide guided teaching and reflection screens for grammatical cases (types AT, AV, and AX).

R29: The system must play the instruction audio and the stimulus audio for a task, and display the associated stimulus and option images.

R30: The system must give immediate feedback, both visual and audible, after each answer, distinguishing a correct answer from an incorrect one.

R31: The system must allow two attempts per trial and must present different feedback for the first incorrect attempt and the second incorrect attempt.

R32: The system must present a remediation (adaption) task to a child whose number of mistakes in the preceding task reaches or exceeds a configured threshold. The default threshold is three mistakes and is configurable per task type.

2.5 Progress, Results, and Navigation

R33: The system must record every student response, storing the trial, the selected option, whether the answer was correct, the attempt number, and the time of the response.

R34: The system must store, for each student and each task, the completion status and the resulting score.

R35: The system must allow a student to resume an unfinished module at the task at which the module was interrupted.

R36: The system must provide a teacher dashboard that summarises the progress of the teacher's classes and students.

R37: The system must allow a teacher to view the results of an individual student for an individual task.

R38: The system must provide menu-based navigation for all users using client-side routing.

R39: The system must present a child-appropriate student interface, using large interactive targets, minimal written text, and audio instructions, so that the interface does not depend on reading ability.

3. Functional Requirements (Should)

R40: The system shall provide a teacher dashboard that displays class-level completion metrics.

R41: The system shall allow results to be filtered and sorted by week, area, class, and student.

R42: The system shall allow an administrator to import a new week of content in the supplied CSV and media format without any change to the application code.

R43: The system shall allow a teacher to reset a module for a student, so that the student can attempt the module again.

R44: The system shall display a confirmation message after each successful teacher or administrator action.

R45: The system shall provide a progress summary that can be printed or exported.

R46: The system shall validate all teacher and administrator input, including the uniqueness of class names, student names, and access codes.

4. Functional Requirements (Can)

R47: The system can record a child's spoken response for speaking tasks, subject to the consent process described in NFR07.

R48: The system can export response data as a CSV file for analysis.

R49: The system can provide gamification elements, such as stars, badges, and a reward screen.

R50: The system can provide a multi-language interface, for example German and English.

5. Non-Functional Requirements

The system processes data about children in Germany. Data protection and system security are therefore mandatory requirements, not optional enhancements.

5.1 Security: Data Protection

NFR01: The system must comply with the General Data Protection Regulation (GDPR), and all personal data must be stored and processed on servers located within the European Union.

NFR02: The system must apply data minimisation. Student records must identify children by a display name or pseudonym and an access code only, and must not store any personal data beyond that which is required to operate the platform.

NFR03: The system must allow a teacher or an administrator to permanently delete a student's account together with all of that student's recorded responses.

NFR04: The system must restrict access to a student's data to that student, the teacher who created the account, and the administrator.

NFR05: The system must record, for each administrative action that creates or deletes an account, the acting user and the time of the action.

NFR06: The system must transmit all data over encrypted connections (HTTPS/TLS).

NFR07: If audio recording of spoken answers is enabled (R47), recordings must be stored encrypted at rest, must be accessible only to the child's teacher and the administrator, and must be deleted together with the student account.

5.2 Security: System Security

NFR08: The system must store all passwords as salted hashes. Passwords must never be stored or transmitted in plain text.

NFR09: The system must authenticate teachers and administrators using signed session tokens, and each token must expire after a defined period of inactivity.

NFR10: The system must generate student access codes that are random and of sufficient length to make guessing impractical, and each access code must be unique across the system.

NFR11: The system must limit the number of consecutive failed login and access-code attempts from the same source, in order to prevent brute-force attacks.

NFR12: The system must validate and sanitise all input received from a client, and must use parameterised database queries, in order to prevent injection attacks.

NFR13: The system must serve media assets and uploaded files only to authenticated sessions that are authorised to access them.

NFR14: The system must not disclose internal error details, stack traces, or database information in any response returned to a client.

NFR15: The system must enforce every access-control rule on the server, and must not rely on the frontend guards alone.

5.3 Usability and Accessibility

NFR16: The student interface must be operable by a child who cannot yet read fluently, through large interactive targets and spoken instructions.

NFR17: The system must be usable on tablets as well as on desktop computers.

NFR18: The system must run on the current versions of the mainstream web browsers.

5.4 Performance and Reliability

NFR19: The system must preload the audio and image assets of a task before the task begins, so that playback is not interrupted while a child is working.

NFR20: The system must retain all responses already recorded when a session is interrupted, and must allow the child to resume the module without loss of those responses.

5.5 Maintainability

NFR21: The task engine must be data-driven, so that new content can be added by importing data without modifying application code.

NFR22: The system must keep learning content separate from learner data, so that content can be updated without affecting recorded responses.

6. Technical and Acceptance Requirements

The following requirements define the delivery technology and the acceptance conditions. They are not counted among the functional requirements.

TR01: The frontend must be implemented in React (JavaScript) with client-side routing provided by React Router.

TR02: Route guards and role guards must be implemented on the frontend as protected route components that wrap the routes they protect.

TR03: The backend must be a Node.js and Express REST API, consumed by the frontend.

TR04: Structured data must be stored in a SQL relational database.

TR05: Authentication must use signed tokens (JWT) for administrators and teachers, and a separate code-based session for students.

TR06: Server-side middleware must verify the token and the role of the requesting user before any protected route handler is executed.

TR07: A content importer must convert the supplied CSV files and media assets into the content format used by the system.

TR08: Media assets and uploaded files must be stored so that they are served only to authorised sessions.

TR09: The core components and services must be covered by automated tests.

TR10: The system must be deployed on infrastructure located within the European Union, in accordance with NFR01.

End of Requirements Specification